

Zhenyao Cui

School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan, China

zycui@hust.edu.cn | czy-trevor.github.io

Research Interests

Transfer learning; multimodal learning; neural decoding and BCIs; foundation models and AI agents.

Education

Huazhong University of Science and Technology

Sep 2024 – Jun 2027

M.Eng. in Intelligent Science and Technology (expected)

- GPA **3.89/4.00**; selected for direct graduate admission through the national recommendation scheme.
- Advisor: Prof. **Dongrui Wu**, IEEE Fellow and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology

Sep 2020 – Jun 2024

B.Eng. in Automation, Elite Engineer Program

- GPA **3.91/4.00**; Outstanding Graduate of HUST.

Publications

[†] equal contribution.

1. **Zhenyao Cui**, Siyang Li, Siyuan Kan, Yudi Wang, Fei Luo, and Dongrui Wu. “A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces.” *IEEE Transactions on Emerging Topics in Computational Intelligence*, Early Access, 2026. doi:10.1109/TETCI.2026.3683713
2. Siyang Li[†], Jiayi Ouyang[†], **Zhenyao Cui**[†], Ziwei Wang, Tianwang Jia, Feng Wan, and Dongrui Wu. “Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces.” *IEEE Journal of Biomedical and Health Informatics*, accepted for publication, 2026. arXiv:2601.07556
3. Dingkun Liu[†], Yuheng Chen[†], Zhu Chen[†], **Zhenyao Cui**, Yaozhi Wen, Jiayu An, Jingwei Luo, and Dongrui Wu. “EEG-FM-Compass: Progress, Benchmarking, and Future Directions for EEG Foundation Models.” *National Science Review*, nwag466, 2026. doi:10.1093/nsr/nwag466
4. **Zhenyao Cui**, Siyuan Kan, Siyang Li, Ziwei Wang, and Dongrui Wu. “SCORE: Subject Coordinate Recovery for Label-Free Cross-Subject EEG-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19134.
5. **Zhenyao Cui**, Siyuan Kan, Dingkun Liu, and Dongrui Wu. “Beyond Trial Averaging: Anchoring Neural and Visual Representations for Few-Repetition Brain-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19128.
6. **Zhenyao Cui**, Jiahui Chai, Siyuan Kan, Yudi Wang, and Dongrui Wu. “MCL-Align: Marginal-Conditional-Label Alignment for Cross-Subject EEG Regression.” Under review, *IEEE Transactions on Biomedical Engineering*, 2026.
7. Siyuan Kan, Huanyu Wu, **Zhenyao Cui**, Fan Huang, Xiaolong Xu, and Dongrui Wu. “CMCRD: Cross-Modal Contrastive Representation Distillation for Emotion Recognition.” *IEEE Transactions on Affective Computing*, 17(2):1837–1846, 2026. doi:10.1109/TAFFC.2026.3664341

Research Experience

Huazhong University of Science and Technology

Apr 2024 – Present

Transfer Learning and Reliable Adaptation

Develop methods that reuse information across individuals and adapt to new-user data, with EEG as the main testbed.

- Proposed **C2R**, a fuzzy-set framework that extends class-conditional transfer learning from classification to continuous EEG regression, and derived a target-domain risk bound based on fuzzy class-conditional probabilities. (*IEEE TETCI, Early Access, 2026*)
- Proposed **MCL-Align** for cross-subject EEG regression, jointly aligning marginal, conditional, and label distributions to account for differences between users. The framework combines Euclidean Alignment, Conditional Direction Matching, and label standardization. (*IEEE TBME, under review*)
- As equal-first author, co-developed **BFT** for lightweight EEG test-time adaptation. Combined knowledge-guided augmentation, Monte Carlo-dropout-inspired feature masking, and a source-trained learning-to-rank module to weight transformed predictions and reduce test-time uncertainty without backpropagation. (*IEEE JBHI, accepted, 2026*)

Research Experience (continued)

Cross-Modal Representation Learning

Study how to align neural and visual representations for retrieval across users and with few repeated recordings.

- Identified systematic coordinate differences between users' neural representations and proposed **SCORE** to recover them through reliable neural–visual landmark matching and orthogonal reorientation, enabling label-free cross-subject EEG-to-image retrieval with frozen encoders. (*under review*)
- Proposed **NEAR** to address non-transitive neural–visual alignment when few repeated recordings are available, using query denoising and image pseudo-anchor prediction to improve brain-to-image retrieval. Evaluated across EEG, MEG, and fMRI. (*under review*)

Foundation Model Evaluation and Neural Data Acquisition

Evaluate pretrained EEG models for continuous prediction and collect neural recordings for new decoding tasks.

- Co-authored **EEG-FM-Compass**; contributed to the literature review and dataset curation, and benchmarked EEG foundation models including **LaBraM** and **CBraMod** for continuous PERCLOS (percentage of eye closure) prediction on SEED-VIG. Compared full fine-tuning with linear probing under cross-subject evaluation and 10% within-subject calibration using RMSE and correlation. (*National Science Review*, 2026)
- Designed and am running a three-stage EEG paradigm spanning image perception, image imagination, and neural speech decoding. Recordings use a 128-channel local high-density array (developed by a collaborator) and a 128-channel whole-head wet-electrode system.

Industry Experience

Alibaba Cloud Intelligence Group

Apr 2026 – Present

Research Intern

- **AutoResearch — Automated ML Experimentation.** Contributed to the development of Alibaba Cloud's internal platform for agent-driven improvement of machine-learning tasks. Task definitions, code branches, and evaluation metrics guide the agent through iterative code modification and evaluation, using results to inform subsequent attempts.
- **Domain-Specific Foundation Embeddings.** Developed a foundation embedding model for Elastic Compute Service (ECS) logs with a domain-specific tokenizer and contrastive pretraining, then paired it with a temporally attentive classifier. On production data, F1 improved from **0.4383** to **0.5306** (+**21.1%** relative).

Dongfeng Motor Group — Multimodal Intelligent Cockpit Evaluation

Jun 2025 – Jun 2026

Industry-funded project — Project Lead

- Led multimodal data collection for intelligent-cockpit evaluation, recording EEG with a 32-channel dry-electrode system and synchronizing it with video, audio, and text to assess users' subjective experience of in-vehicle functions.

Selected Projects and Competitions

SSVEP-Controlled Robotic Arm with Adaptive Step Size

Sep 2025 – Dec 2025

- Built a 32-channel wet-electrode SSVEP acquisition and decoding system for precise robotic-arm control, using position-aware adaptive step sizing to balance speed and endpoint precision. It won the **national championship** in the Brain-Controlled Robotic Arm track of the 2025 China Brain-Computer Interface Competition. The system was later deployed at Wuhan Optics Valley Biotech City; the championship achievement was covered by China Central Television and *Global Times*.

Porcine ECoG-Based Continuous Joint-Motion Decoding

2024

- Developed decoding algorithms for continuous porcine joint-motion trajectories from ECoG, using segment-slicing augmentation tailored to recurring motion patterns during running. The project won **First Prize** in the minimally invasive BCI track at the 2024 World Robot Contest Championship.

Selected Awards and Honors

- **National Champion**, China Brain-Computer Interface Competition, Brain-Controlled Robotic Arm track, 2025.
- **First Prize**, Porcine ECoG-Based Continuous Joint-Motion Decoding, World Robot Contest Championship, 2024.
- **First Prize**, National Undergraduate Intelligent Vehicle Competition, Multi-Vehicle Formation track, 2022.
- First-Class Graduate Academic Scholarship, HUST, 2024–25 and 2025–26; Yunnan Scholarship, 2024–25; Huawei Scholarship, 2021–22; Outstanding Graduate of HUST, 2024.